

Chih-Kai Wang

Software Engineering & AI Systems Intern | Python, TypeScript, Autonomous Agents

[Email](#) | [GitHub](#) | [Portfolio](#) | Taipei, Taiwan

PROFILE

I design autonomous agent runtimes, sandboxed execution environments, and formal verification tools using Python and TypeScript. Seeking software engineering and AI systems internships, with a focus on verifiable execution, sandboxing, and developer tooling.

EDUCATION

National Taipei University of Education | Expected 2028

B.S. student, Mathematics Division, Department of Mathematics and Information Education.

SELECTED PROJECTS

[DSH Architecture Lab](#) | Node.js, Linux VM, DeepSeek Harness

- Architected a headless agent research runtime with Lima Linux VM isolation, micro-cent token broker metering, and immutable cryptographic protocol receipts.
- Executed factorial empirical trials across memory/planning recipes; achieved 100% external test pass rate with DeepSeek-V4.1-Flash.

[RuleShift](#) | Python, Apple MLX, Gymnasium, Preact

- Developed a deterministic testbed evaluating agent memory adaptation under shifting rules across 800 paired tasks with automated state verification and replayable evidence.
- Empirically showed on-demand retrieval matches selective updating (95.6% success) with fewer tool calls and lower compute cost.

[HonestCI](#) | TypeScript, Node.js, GitHub Actions

- Built a developer-first CLI and GitHub Action verifying fresh JUnit XML evidence, test-count retention, and false-green CI prevention. Published v1.0.4 on npm.

[RigorGraph](#) | Python, JSON Schema, HTML Reports

- Implemented local-first claim-evidence DAGs, four-eyes review verification, SHA-256 byte fingerprints, and deterministic offline HTML audit dossiers.

RESEARCH & OPEN SOURCE

[ProofWeave Core](#): structured mathematical claim certification with Lean 4. **[TokenScope](#)**: interactive causal attention and BPE tokenizer lab. **[Second Agent Kit](#)**: macOS Seatbelt sandbox and memory isolation.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript (Node.js 20+), Lean 4, SQL, Bash

Systems & Tools: Linux VM, Docker, macOS Seatbelt, Git, GitHub Actions, DeepSeek Harness, WebMCP, PyTorch, MLX

Last updated: September 2026